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viii) To understand the practical application of prev and next pointers.

4] Algorithms

Algorithm 1: Add a new patient
Step 1: Start
Step 2: Create a newNode using dynamic memory allocation
Step 3: Enter patient ID, Name, Age, and Disease
Step 4: Set newNode → prev = NULL and newNode → next = NULL
Step 5: Check whether head = NULL.
- if yes, make head = newNode.
- otherwise, traverse the list to the last node

Step 6: Connect the new node with the last node:
- last → next = newnode
- newNode → prev = last

Step 7: Display "patient added successfully"

Step 8: Stop

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Algorithm 2: Display patient records

Step 1: Start
Step 2: check whether head == NULL
- if yes, display "No patient record found".
- Otherwise, continue.

Step 3: Set temp = head.
Step 4: Display the patient information.
Step 5: Move to the next node using temp = temp -> next.
Step 6: Repeat until temp == NULL
Step 7: Stop

Algorithm 3: Search patient
Step 1: Start
Step 2: ~~Check whether~~ head == NULL.
- if

Step 2: Enter the patient ID to be searched.
Step 3: Set temp = head.
Step 4: compare temp -> patient ID with entered ID
- if equal. display the patient details and stop searching.

- otherwise, move to the next node.

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Algorithm 4: Delete patient After Discharge.

Step 1 : Start.

Step 2 : Enter the patient Id to be deleted

Step 3 : Search for the node containing that patient ID.

Step 4 : if the patient is not found, display patient not found.

Step 5 : if the node is the first node update head.

Step 6 : Otherwise, connect the previous node to the next node.

temp → prev → next = temp → next.

Step 7 : if a next node exists, update its previous pointer.

temp → next → prev = temp → prev

Step 8 : Delete the node using delete temp.

Step 9 : Display "patient discharged successfully"

Step 10 : Stop